

ALFRED LEIGH

Aspiring Aerospace Engineer

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PROFILE

Year 12 student (A-level Mathematics, Further Mathematics, Physics and Computer Science) committed to a career in aerospace engineering. I design and build real hardware from first principles: currently a from-scratch quadcopter and a 72V electric enduro motorcycle, each documented to industry conventions (product design specifications, trade studies, engineering calculations, failure logs) and public on GitHub. Completed a week of civil engineering work experience with Berkeley Group on live London sites in summer 2026. Seeking aerospace placements and competition opportunities.

ENGINEERING PROJECTS

First-Principles Quadcopter • github.com/a365l/quadcopter-project Jun 2026 – present

- Wrote a Product Design Specification with 15 quantified requirements (3:1 thrust-to-weight, arm resonance $\geq 1.5\times$ motor frequency, ≤ 3.5 kg all-up weight) before touching CAD.
- Building a first-principles calculation set (momentum-theory propulsion sizing, Euler-Bernoulli cantilever resonance, bolt preload), each to be validated against thrust-stand and flight-test measurements.
- Pressure-tested feasibility with researchers at Imperial College London; £700 budget, 10 weeks to first flight.

72V Enduro Electric Motorcycle • github.com/a365l/enduro-emotorcycle-build 2025 – present

- Built a full electric enduro motorcycle from a bare frame across three iterations, culminating in a QS205 hub motor and Fardriver ND72450 controller (200 A continuous / 450 A peak, 32.4 kW peak electrical); every phase photo-documented on GitHub.
- Predicted ~ 55 mph top speed and ~ 3.6 s 0–30 mph from first principles; maintain a failure log (symptom $\rightarrow$ root cause $\rightarrow$ fix $\rightarrow$ retest) and test-vs-prediction records for every subsystem.
- Diagnosed a dead-on-arrival controller (shorted MOSFETs) with a 10-second continuity check; designed a custom throttle adapter harness; $\sim$ £3,000 build part-funded by stripping and flipping donor bikes.

Roboscan • RF security multi-tool: sub-GHz/NFC/IR interface device (CC1101, ESP32/Pi Pico) plus network-enumeration tooling; first major embedded build.

WORK EXPERIENCE

Berkeley Group • civil engineering placement, Bermondsey Place & Trent Park, London Jun–Jul 2026

- One-week placement across two live sites, shadowing site engineers on temporary works, substructure, piling and setting out, and working alongside quantity surveyors, planners and subcontractors.
- Hands-on with a Leica total station and rotating laser level for setting out and level checks; observed concrete QC catch a failed slump test on a live pour, alongside 7- and 28-day cube testing.
- Worked through manhole formation-level calculations and BS 8666 rebar shape codes on site; wrote the week up as a 20-slide technical report.

Kras Carpentry • paid commercial work 2024 – present

- Hands-on carpentry with power tools, basic household plumbing and electrical installation; designed, built and maintain the company website, krascarpentry.co.uk.

Simply Exports • family plant & machinery export business

- Grew up working in the yard: combustion engine repair, stripping plant machinery and general workshop work from a young age; built the company website, simplyexports.co.uk.

EDUCATION

Sixth Form A-levels (from Sept 2026): Mathematics, Further Mathematics, Physics, Computer Science

Debden Park High School • GCSEs, predicted (results Aug 2026): Computer Science 9, Geography 9, Triple Science 8/8/9, Mathematics, English 2021 – 2026

SKILLS

- **Engineering practice:** product design specifications, trade studies, first-principles calculation, failure logging, test-vs-prediction validation, BOM & budget management
- **Electronics:** soldering & micro-soldering, crimped vehicle looms, 72V high-voltage battery systems, BLDC motor control (FOC), embedded systems (ESP32, Pi Pico, Raspberry Pi)
- **Software:** C++, Python, Kotlin, JavaScript/React, Linux (Debian/Arch) administration & networking; practical cyber-security (Kali, TryHackMe)

LANGUAGES & INTERESTS

English (native), Slovak (fluent), Czech (fluent) • Sailing (RYA Level 4) • Bouldering (V6) • Boxing & Brazilian Jiu-Jitsu • Competitive swimming • Guitar (8 years)